BIOL-1 Practice Test 03 — Answer Sheet

Exam 02 Preparation (Modules 7–11)

Name:		Date:		Score:	

✂️ **TEAR-OFF ANSWER SHEET:** record every Part A: Multiple Choice (25 questions) and Part B: Fill in the Blank (10 questions) answer on this page. Free response begins on the reverse side. Detach and turn in this sheet.

Part A: Multiple Choice (25 questions)

1.		**10.**		**19.**	
2.		**11.**		**20.**	
3.		**12.**		**21.**	
4.		**13.**		**22.**	
5.		**14.**		**23.**	
6.		**15.**		**24.**	
7.		**16.**		**25.**	
8.		**17.**			
9.		**18.**			

Part B: Fill in the Blank (10 questions)

1.		**6.**	
2.		**7.**	
3.		**8.**	
4.		**9.**	
5.		**10.**	

BIOL-1 Practice Test 03 — Part C: Short Answer (4 questions)

Free response. Write clearly in the lined spaces.

1. **(Module 7)** Write the complementary DNA sequence AND the transcribed mRNA sequence for the following DNA template strand: **3′-TAC GGG TTA ACG ACT-5′**.
2. **(Module 8 & 9)** Explain the difference between Mitosis and Meiosis in terms of their overall purpose for a multicellular organism like a human. Briefly link how Meiosis connects to Mendel's Law of Segregation.
3. **(Module 10)** Describe how it is possible for a muscle cell and a nerve cell in your body to look and function so differently, despite containing the exact same DNA.
4. **(Module 11)** Choose ONE biotechnology tool discussed in class (e.g., PCR, Gel Electrophoresis, CRISPR, Recombinant Plasmids). Explain what it does and provide one real-world application or example of its use.

Response 1 — Question Number: ______

Response 2 — Question Number: ______

Response 3 — Question Number: ______

Response 4 — Question Number: ______

BIOL-1 Practice Test 03 — Question Booklet

Answer Parts A and B on the tear-off sheet. Keep this booklet with your exam materials.

Part A: Multiple Choice (25 questions)

Choose the best answer for each question.

Module 7: Molecular Genetics

1. The enzyme that unwinds the DNA double helix during replication is: A) RNA polymerase B) DNA polymerase C) Helicase D) Ligase

2. Okazaki fragments are synthesized on the: A) Lagging strand B) mRNA transcript C) Leading strand D) tRNA molecule

3. Transcription is the process of converting: A) RNA into a protein B) DNA into RNA C) DNA into a new DNA molecule D) Protein into an observable trait

4. Which part of a primary mRNA transcript is removed during RNA splicing before it leaves the nucleus? A) Introns B) Promoters C) Codons D) Exons

5. A mutation that shifts the entire reading frame of the ribosome is called a: A) Missense mutation B) Silent mutation C) Frameshift mutation D) Point mutation

Module 8: Cellular Genetics (Mitosis & Meiosis)

6. Which phase of the cell cycle is characterized by DNA replication? A) G2 phase B) M phase C) S phase D) G1 phase

7. After a single human cell undergoes meiosis, the resulting cells are: A) 4 haploid, genetically distinct cells B) 2 diploid, genetically identical cells C) 4 diploid, genetically distinct cells D) 2 haploid, genetically identical cells

8. Crossing over, which increases genetic diversity, occurs during: A) Metaphase II of Meiosis B) Prophase of Mitosis C) Prophase I of Meiosis D) Anaphase I of Meiosis

9. Nondisjunction during meiosis can lead to conditions such as Down syndrome, which is caused by trisomy of chromosome: A) 18 B) 21 C) 23 D) Y

10. The p53 protein is heavily involved in: A) Forming the cleavage furrow B) Breaking down the nuclear envelope C) Stopping the cell cycle if DNA is damaged D) Attaching spindle fibers to kinetochores

Module 9: Inheritance Genetics

11. The physical appearance or observable trait of an organism is its: A) Genotype B) Phenotype C) Karyotype D) Allele

12. In a monohybrid cross between two heterozygous parents (Aa × Aa), what is the expected phenotypic ratio for a completely dominant trait? A) 1:2:1 B) 3:1 C) 9:3:3:1 D) 1:1

13. A true-breeding red snapdragon is crossed with a true-breeding white snapdragon, producing offspring that are all pink. This is an example of: A) Incomplete dominance B) Polygenic inheritance C) Sex-linked inheritance D) Codominance

14. Human blood types exhibit which type(s) of inheritance pattern(s)? A) Sex-linked recessive B) Simple dominance only C) Incomplete dominance D) Codominance and multiple alleles

15. Why are sex-linked recessive traits more common in human males than females? A) Males have a stronger Y chromosome. B) The X chromosome in males is larger than in females. C) Males only have one X chromosome, so a single recessive allele determines the phenotype. D) Females lack the genes required for these traits entirely.

Module 10: Epigenetics

16. Epigenetics refers to changes in gene expression that: A) Only occur in prokaryotes. B) Do not change the DNA sequence but can be passed to daughter cells. C) Alter the underlying DNA sequence permanently. D) Are only caused by environmental toxins.

17. DNA methylation generally has what effect on gene expression? A) It increases transcription. B) It causes the DNA to mutate rapidly. C) It has no effect on transcription. D) It decreases or silences transcription.

18. The lac operon in E. coli allows the bacteria to produce enzymes to digest lactose only when lactose is present. This is an example of: A) A sex-linked trait B) A frameshift mutation C) Epigenetic inheritance D) Gene regulation

19. Acetylation of histones usually results in: A) Loosely packed chromatin and increased transcription. B) Immediate degradation of the DNA. C) Tightly packed chromatin and decreased transcription. D) The splicing of introns out of mRNA.

20. The random inactivation of one X chromosome in female mammalian cells results in physical traits like tortoiseshell coloring in cats. The inactivated X chromosome condenses into a structure known as a: A) Centrosome B) Barr body C) Telomere D) Nucleosome

Module 11: Genomics & Biotechnology

21. The primary function of restriction enzymes in biotechnology is to: A) Amplify DNA sequences. B) Analyze protein folding. C) Glue Okazaki fragments together. D) Cut DNA at highly specific sequence sites.

22. Polymerase Chain Reaction (PCR) is used to: A) Rapidly make millions of copies of a specific DNA segment. B) Translate mRNA into a polypeptide chain in a test tube. C) Separate DNA fragments by size. D) Insert foreign DNA into a bacterial plasmid.

23. During gel electrophoresis, DNA fragments move toward the positive electrode because: A) DNA is positively charged. B) DNA has a neutral charge but is pulled by gravity. C) agarose gel attracts DNA chemically. D) the phosphate backbone gives DNA a negative charge.

24. The CRISPR-Cas9 system was originally discovered in: A) Bacteria as a rudimentary cellular immune system against viral bacteriophages. B) Plants to resist fungal infections. C) Human liver cells to repair damaged DNA. D) Viruses as an attack mechanism against hosts.

25. Transgenic organisms, such as bacteria engineered to produce human insulin, are possible because: A) Transcription only works in eukaryotic laboratory cells. B) Bacteria and humans have identical DNA sequences. C) Human genes naturally mutate into bacterial genes. D) All organisms share a nearly universal genetic code.

Part B: Fill in the Blank (10 questions)

Word Bank: Biotechnology, Cytokinesis, Epigenetic, Heterozygous, Lethal, mRNA, Operon, Smaller, Trisomy, Uracil

26. In RNA, the nitrogenous base ____ replaces the Thymine found in DNA.

27. According to the Central Dogma of molecular biology, DNA is transcribed into ____, which is then translated into a protein.

28. The division of the cytoplasm at the very end of mitosis or meiosis is called ____.

29. The presence of three copies of a chromosome instead of the normal two is referred to as a ____.

30. A gene variant that causes death when inherited in homozygous form is known as a ____ allele.

31. An individual with two different alleles for a specific trait (e.g., Bb) is called ____.

32. In gene regulation, a(n) ____ is a cluster of genes under the control of a single promoter, commonly found in prokaryotes.

33. Identical twins have the exact same DNA sequence, but as they age, differences in their gene expression accumulate due to ____ changes.

34. In gel electrophoresis, ____ fragments of DNA travel through the gel matrix faster and farther than larger fragments.

35. The use of an organism, or a component of an organism, to make a product or process for human use is known as ____.
